

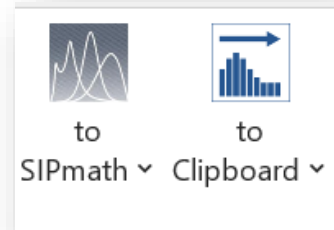
Getting Started With

[Click Here For Contents](#)

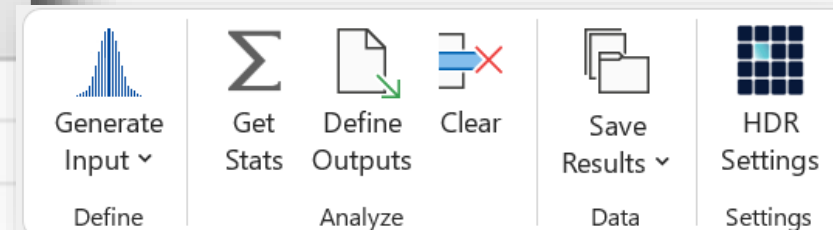


If you get a “Compile error in hidden module”, you may need to enable ActiveX controls.

See Appendix

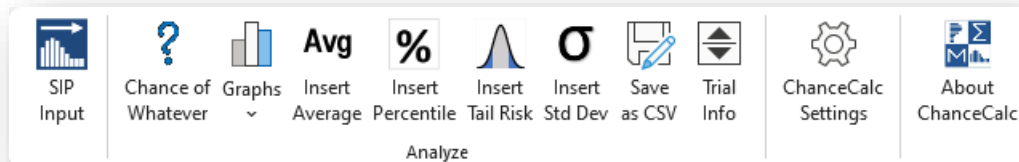


Monte Carlo



Contents

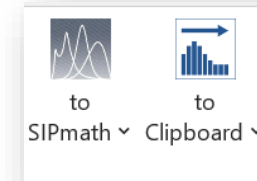
1. [Introduction](#)
2. [ChanceCalc Ribbon](#)



Basic Commands

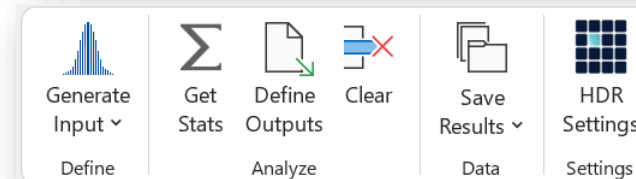
- a) SIP Input
- b) Graphs
- c) Insert Average, Chance of Whatever, Percentile, Tail Risk & Standard Deviation
- d) Save as CSV
- e) Trial Info
- f) Settings: Simulation, Formatting, Diagnostic

3. [Metalog Ribbon](#)



- a) Select mode
 - i. SIPmath 3.0
 - 1) Sub-mode “to Clipboard”
 - 2) Sub-mode “to File”
 - ii. Excel
 - 1) Sub-mode “Paste to Excel”
 - 2) Sub-mode “Link to Excel”

4. [Monte Carlo Ribbon](#)



- a) Input distributions
- b) Clear output
- c) Get Stats
- d) Define Output
- e) Save Results
- f) HDR Settings



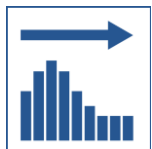


1. Introduction

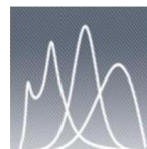
ChanceCalc is an Excel add-in for creating chance-informed dashboards in your spreadsheet based on uncertainties stored in **Stochastic Libraries** as CSV files, or in the open SIPmath™ 2.0 or 3.0 Standard. The models created with ChanceCalc are stand-alone Excel files that perform hundreds or thousands of simulation trials per key-stroke and do not require the add-in to run. It includes a separate tab for importing Metalog distributions for Tom Keelin's Metalog Templates.

ChanceCalc Monte Carlo expands the capabilities to include Monte Carlo simulation with internal generation of random variates and the ability to save stochastic libraries as SIPmath 2.0 files, including metadata.

The SIPmath™ 3.0 Standard is based on Tom Keelin's revolutionary **Metalog distribution** and is driven by Doug Hubbard's **HDR portable random number generator**. Streams of up to 100 million trials of a very wide variety of probability distribution may be encoded into 18 Metalog parameters and 4 HDR seeds. The open SIPmath 3.0 Standard wraps these 22 numbers in a JSON object, which can generate identical streams of random variates in any computer environment.



[The SIPmath 3.0 Standard](#)

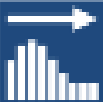


[The Metalog Distribution](#)



[The HDR Generator](#)

**a. Basic
Commands**



SIP
Input



Chance of
Whatever



Graphs
~



Avg
Insert
Average



%
Insert
Percentile



Insert
Tail Risk



σ
Insert
Std Dev



Save
as CSV



Trial
Info



ChanceCalc
Settings



About
ChanceCalc

Determine the chance that any formula is greater or less than any specified target

Determine the simulated average of any formula

Determine the simulated average of results beyond a specified target

Trial Info

Settings, including number of trials & Chance percent rounding

Select SIP Libraries and input into your model

Create histograms or cumulative charts of any formula or scatter plots of any pair of formulas

Determine a specified percentile of any formula

Standard Deviation

Save results as CSV file

Check version number



SIP
Input

a) SIP Input



Libraries types supported are CSV, SIPmath™ 2.0 and 3.0 Standard and may be accessed from the web or your computer.

1. Select destination cells.
It saves time to do this
before clicking SIP Input

2a. Access a copied SIP library after right
clicking and copying Link from online library

2c. Create SIPs with
Excel formulas (CC
Monte Carlo only)

2b. Open a SIP
library from your
local machine

3. Select
SIPs to Insert

Optionally insert
SIP names

Optionally specify
Sparkline location

Insert SIPs

Destination
C1

Library
Gaussian Copula Development Time Library

Local Lib... Clipboard Monte Carlo

Filter

☐ Sort alphabetically ☒ Select All

Accounts
Products
Orders
Fulfillment
Marketing

☐ Insert Name To Left

Sparkline Location
C1

OK Cancel

- ☐ Open link in new tab
- ☐ Open link in new window
- ☐ Open link in InPrivate window

Product

Demand for a product

Flight Hours

Part failure rate

Number and

Development

Development

Covid-19 Fc

Hospitalizati

Flight System Failures

Distribution of flight hours before failure of hypothetical aircraft systems

Save link as

Copy link

Add to Collections

Share

Web select

Ctrl+Shift+X

Web capture

Ctrl+Shift+S

Inspect



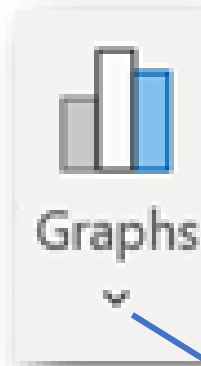
b) Graphs

1. Select formula cell to graph before clicking button

2. Click button to create histograms, cumulative graphs

3. You may optionally name formulas when you graph them. This is useful if you plan to save your outputs.

4. Click down arrow for scatter plots



☐ Chart Selected
☒ Chart SIPmath Out

Names (optional)

Outputs to Graph

Number of Bins: 50

☐ Sparkline ☐ Integer Charts?
☒ Excel chart
☐ Chart data only

Histogram Starting Location

Cumulative Chart Starting Location

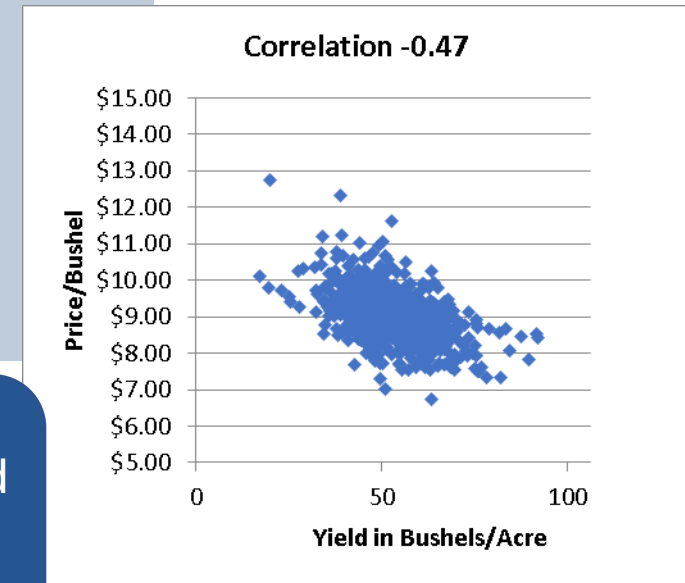
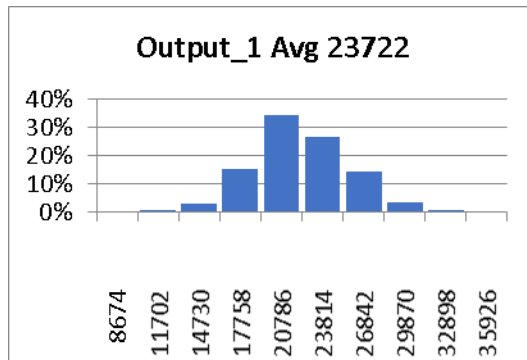
☒ Put multiple charts in rows
☐ Put multiple charts in columns

X output cell

Y output cell

Optional Correl Cell

NOTE: the model will be automatically initialized, and outputs automatically defined, if necessary to make a graph.





Chance of
Whatever



c) Chance of Whatever

1. Select location of
Chance cell before
clicking button

2. Fill in fields and
click **OK**

NOTE: ALL Stats
button on this ribbon
will automatically
initialize the model,
and automatically
define formula cells
as outputs, if needed
to make a graph.

3. You may optionally name formulas
when you get statistics on them. This is
useful if you plan to save your outputs.

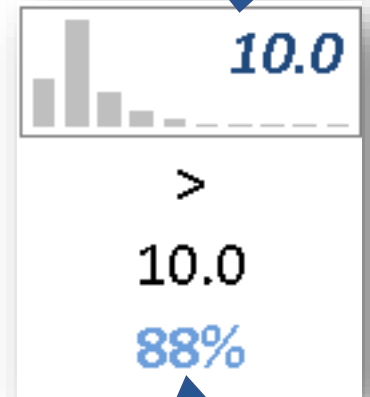
Formula will be simulated
with a resulting sparkline
histogram. Text will be
Italics to indicate a single
trial of many.

Chance of Whatever

Names (optional)	Total_Duration
Formula	Sheet1!\$D\$8
Relation (e.g. < or >=)	Sheet1!\$D\$9
Target value	Sheet1!\$D\$10
Stats Destination	Sheet1!\$D\$11
Sparkline Destination	D8

OK **Cancel**

4. The optional Sparkline Destination will be
auto-filled based on the Formula address and
the Default Sparkline Position in the Settings
dialog. You may also manually adjust it.



Chance appears in
Blue to indicate a
statistic calculated
over all trials.

c) contd. Insert Avg, Standard Deviation, & Percentile



1. Select location of Stats before clicking button

Reminder: You may optionally name formulas when you get statistics on them. This is useful if you plan to save your outputs.

Average / Standard Deviation (same syntax)

Avg
Insert Average

Names (optional)

Average of

σ
Insert Std Dev

Stats Destination

Sparkline Destination

OK Cancel

Percentile

%
Insert Percentile

Names (optional)

Output

Desired %

Stats Destination

Sparkline Destination

OK Cancel

2. Select Formula you want the Stats for

4. The optional Sparkline Destination will be auto-filled based on the Formula address and the Default Sparkline Position in the Settings Dialog. You may also manually adjust it.

Stats appears in blue to indicate a statistic calculated over all trials.

3. Select desired percent



c) contd. Insert Tail Risk

(The Average Above or Below a Specified Value)



1. Select location of Tail Risk result before clicking button

The 'Tail Risk' dialog box contains the following fields and buttons:

- Names (optional)**: An empty text box.
- Tail Risk of**: A text box containing 'Sheet1!\$D\$6'.
- Relation (e.g. >= or <)**: A text box containing 'Sheet1!\$E\$6'.
- Tail Value**: A text box containing 'Sheet1!\$F\$6'.
- Stats Destination**: A text box containing '\$D\$7'.
- Sparkline Destination**: A text box containing 'D6'.
- OK** and **Cancel** buttons at the bottom.

Callouts from the numbered instructions point to the following fields:

- Callout 1 points to the **Stats Destination** field.
- Callout 2 points to the **Tail Risk of** field.
- Callout 3 points to the **Relation** field.
- Callout 4 points to the **Tail Value** field.
- Callout 5 points to the **Sparkline Destination** field.

2. Select Formula you want the Tail Risk for

3. Relation may be typed directly or refer to a cell or range

4. Desired tail value may be typed directly or refer to a cell or range.

5. The optional Sparkline Destination will be auto-filled based on the Formula address and the Default Sparkline Position in the Settings Dialog. You may also manually adjust it.

Reminder: You may optionally name formulas when you get statistics on them. This is useful if you plan to save your outputs.

Stats appears in blue to indicate a statistic calculated over all trials.

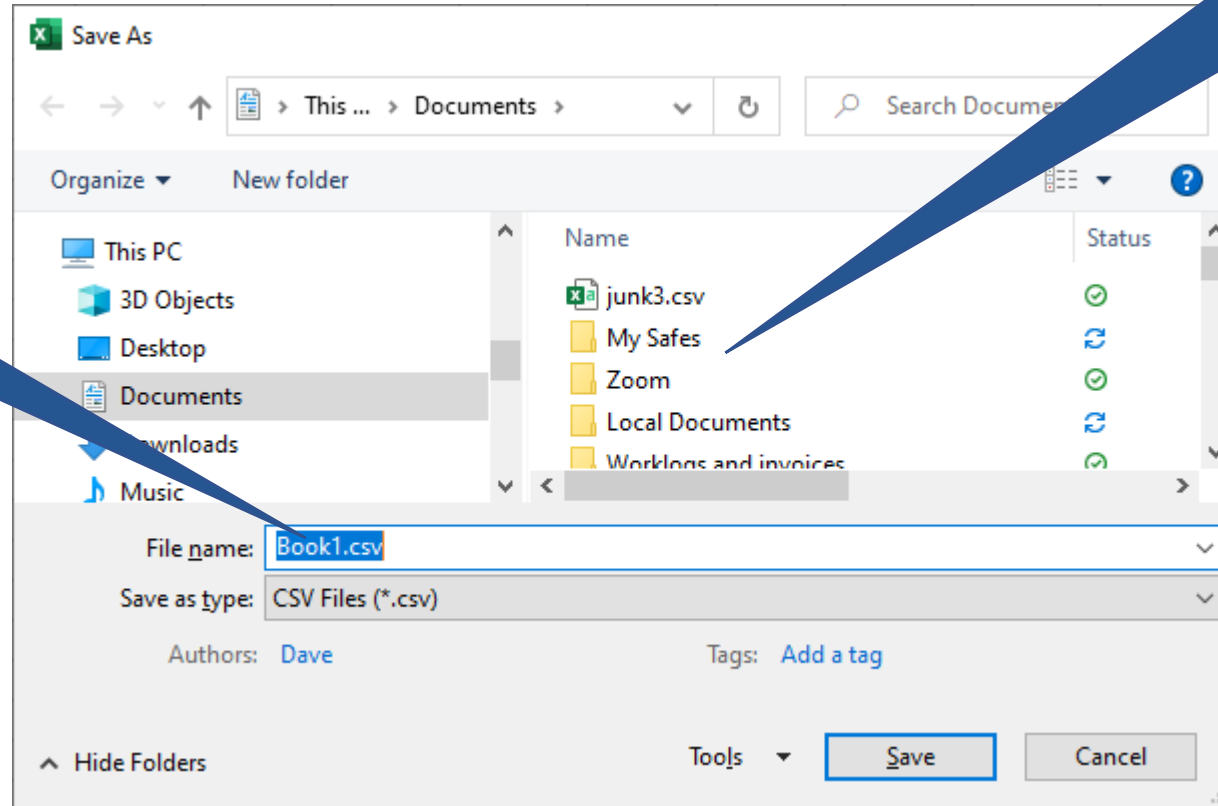


d) Save as CSV

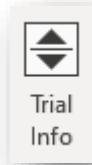


1. Enter file name under which to save data

2. Select folder to save data



Simulated Data from the PMTable Tab will be saved with the name you specify



e) Trial Info

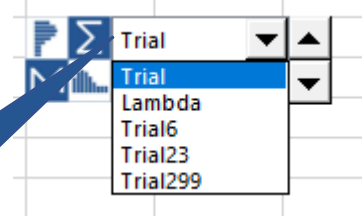
NOTE: You may specify the control to be automatically inserted upon model initialization from the Settings dialog.



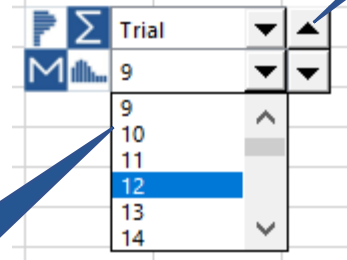
'Floating' controls that will be placed when pressing Trial Info button.



First drop-down selects display of either Trial numbers or Metadata values.



Second drop-down selects Trial number to display.



Spin button also changes trial number.



Trial Info control can be dragged to any desired position by clicking on the Logo.



f) Settings - Simulation

**For Formatting and
Diagnostic Settings**

[Click Here](#)

Limit precision on
Chance of Whatever to
avoid over specificity.

Check to automatically
insert Metadata Trial Info
Controls box.

Default location for
Metadata Trial Info
Controls box.

Settings

Simulation | Formatting | Diagnostic

Number Of Trials 1000

Default Hist. Bins <=100 10

Current Var Id

☐ Round Chance of Whatever to...
5 percent

☒ Metadata Box to Appear at Initialization

Put Metadata controls at... D2

Advanced Settings

OK Cancel

Display number of trials
(3.0 Libraries only). 2.0
Libraries are pre-
specified.

Default number of bins in
a histogram.

Default Var ID value for
HDR random numbers

f) Settings – Formatting & Diagnostic

Settings

Simulation **Formatting** Diagnostic

☐ Use Palette Formatting

Open Palette

Default Sparkline Location

☐ Above

☐ Left ☒ Formula Cell ☐ Right

☐ Below

☐ No sparkline

Advanced Settings

OK Cancel

Get formatting from the Palette file. Change formats in the Palette to control ChanceCalc formats.

Open Palette file for editing.

Set default position of sparklines as relative to formula cells used in SIP Input, Chance of Whatever, Avg, Percentile, and Tail Risk.

Settings

Simulation **Formatting** Diagnostic

☐ Hide Most SIPmath ranges

☒ Turn on Auto Calc in Every Command

☒ Warn when turning on AutoCalc

☒ Show SPT Random Cell Suggestion

Advanced Settings

OK Cancel

Hides internal names so your own defined ranges are easier to identify in the Name Manager

Turns on AutoCalc if it was turned off by mistake.

Offers advice when creating SPT random cells

Warn you if AutoCalc is being turned on

Allows creating and switching between multiple PMTable sheets with differing names.

f) Settings – Advanced Settings

Settings

Simulation | Formatting | Diagnostic

Number Of Trials: 1000

Default Hist. Bins ≤ 100 : 10

Current Var Id:

☐ Round Chance of Whatever to... 5 percent

☒ Metadata Box to Appear at Initialization

Put Metadata controls at... A1

☒ Show Metalog Tab

☒ Show Monte Carlo Tab

☐ Enable Multiple PMTables

Advanced Settings

OK Cancel

Press the Advanced Settings button to show the Advanced Settings options.

Check the Show Metalog Tab checkbox to show the Metalog section of the Ribbon

Check the Show Monte Carlo Tab checkbox to show the Monte Carlo section of the Ribbon

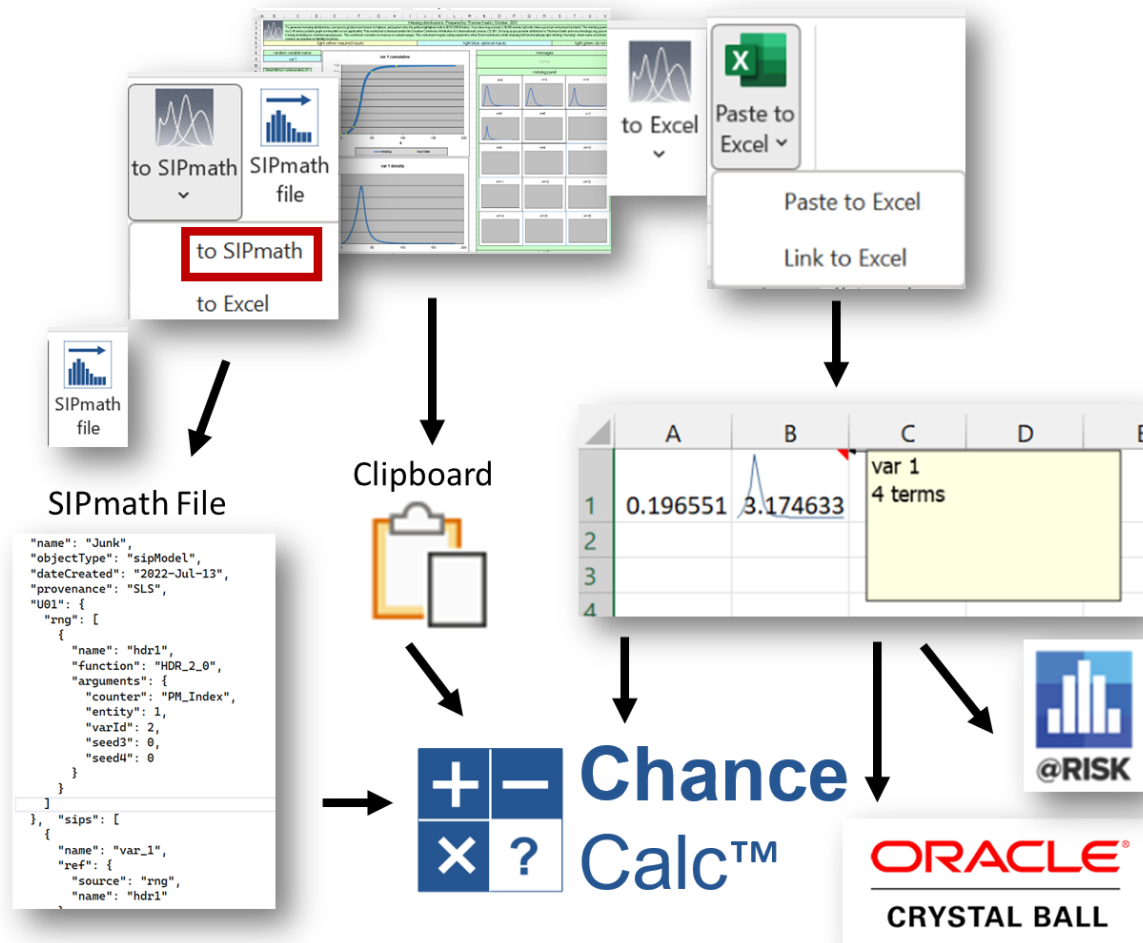
Check the Enable Multiple PMTables checkbox to enable multiple PMTable pages in your model.

Not recommended for beginners.

3. Metalog Ribbon



Metalog Template



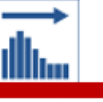
3. Metalog Interface



SIPmath and Excel Modes

The interface has two modes with two sub modes as shown in the table below.

Modes Summary

SIPmath Mode		Excel Mode	
 to SIPmath ▾	 to Clipboard ▾	 to Excel ▾	 Paste to Excel ▾
Creates 3.0 Standard JSON stochastic libraries, either copied to the clipboard or as JSON files.		Creates Metalog formulas to be either pasted or linked directly to your spreadsheet model.	
to SIPmath to Excel		to SIPmath to Excel Paste to Excel Link to Excel	
 to Clipboard ▾		 Paste to Excel ▾	
Click to specify Library Name, Provenance and HDR Seeds. In your Excel model from the ChanceCalc menu, use Clipboard option from the SIP Input Command.		In your Excel model, click to specify location of Random cell, HDR seeds and optional graph and density data. Metalog formula will be pasted into the model.	
 to File ▾		 Link to Excel ▾	
Click to specify Library Name, Provenance and HDR Seeds. You will be prompted to save the JSON SIPmath file to your desktop.		In your Excel model, click to specify location of Random cell, HDR seeds and optional graph and density data. Also specify range for cells linked to parameters in the Metalog template.	

3. Metalog Interface



1. Load a metalog template file

2. Click the Metalog tab

3. Enter the data for your distribution

4. Select the desired boundary type

5. Select the desired boundary values

7. Select "to Excel" mode or "to SIPmath" mode

6. Specify the desired number of terms

The screenshot shows the Metalog Excel interface. The ribbon includes tabs for File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Developer, Add-ins, Help, Script Lab, ChanceCalc, Metalog, and Monte Carlo. The Metalog tab is active, displaying a worksheet with various input fields and a plot. The worksheet contains a title 'Metalog distributions. Prepared by Thomas Keelin, October, 2021.' and a detailed instruction block. Below this, there are sections for 'random variable name', 'var 1', 'bounds', 'metallog specifications', 'regularization', and 'input parameters'. A central plot titled 'var 1 cumulative' shows a cumulative distribution function. To the right, a 'messages' section shows '(none)'. Below the messages, a 'metalog panel' displays a grid of probability density function plots for different numbers of terms (n=5 to n=13). The 'to Excel' mode is selected in the bottom right corner.

random variable name	var 1

bounds (u = unbounded, sl = semibounded lower, su = semibounded upper, b = bounded).
u

metallog specifications	m	n terms
parameters (2-10000)	5	5

regularization	(0,1,...,10)	0

input parameters	y	x
	0.900	85.0
	0.010	8.0
	0.900	50.0
	0.100	20.0
	0.500	30.0

var 1 cumulative

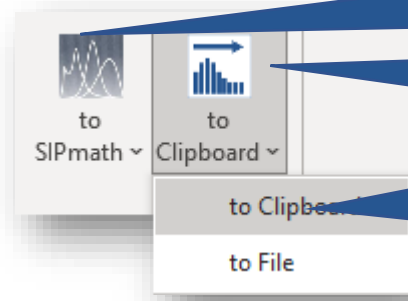
messages: (none)

metalog panel

n=5, n=6, n=7, n=8, n=9, n=10, n=11, n=12, n=13

to Excel mode or "to SIPmath" mode

SIPmath mode, Clipboard sub-mode



Button images and text will change depending on the last mode and sub-mode chosen

Use SIP Input after copying a library to the clipboard



Create a single-SIP library and copy it to the clipboard for use in the SIP Input dialog

Press Clipboard to fill dialog with clipboard library data

Library Data

Library Name	Steelhead
Library Provenance	SLS
SIP Name	Weight
Entity	1
VarId	2
Seed3	
Seed4	

OK Cancel

Name of the library

Source of the data

Name of the SIP within the library

Seeds for the HDR random formula used by the Metalog formula to create the library data values.

Enter text without quotes in Seed3 or Seed4 for user specified seed.

Insert SIPs

Destination: C1

Library: Steelhead

Local Lib... Clipboard Monte Carlo

Filter:

☐ Sort alphabetically

Weight

☐ Insert Name To Left

Sparkline Location: C1

OK Cancel

Make sure to select a SIP

Press OK to insert the Metalog library formula in Excel

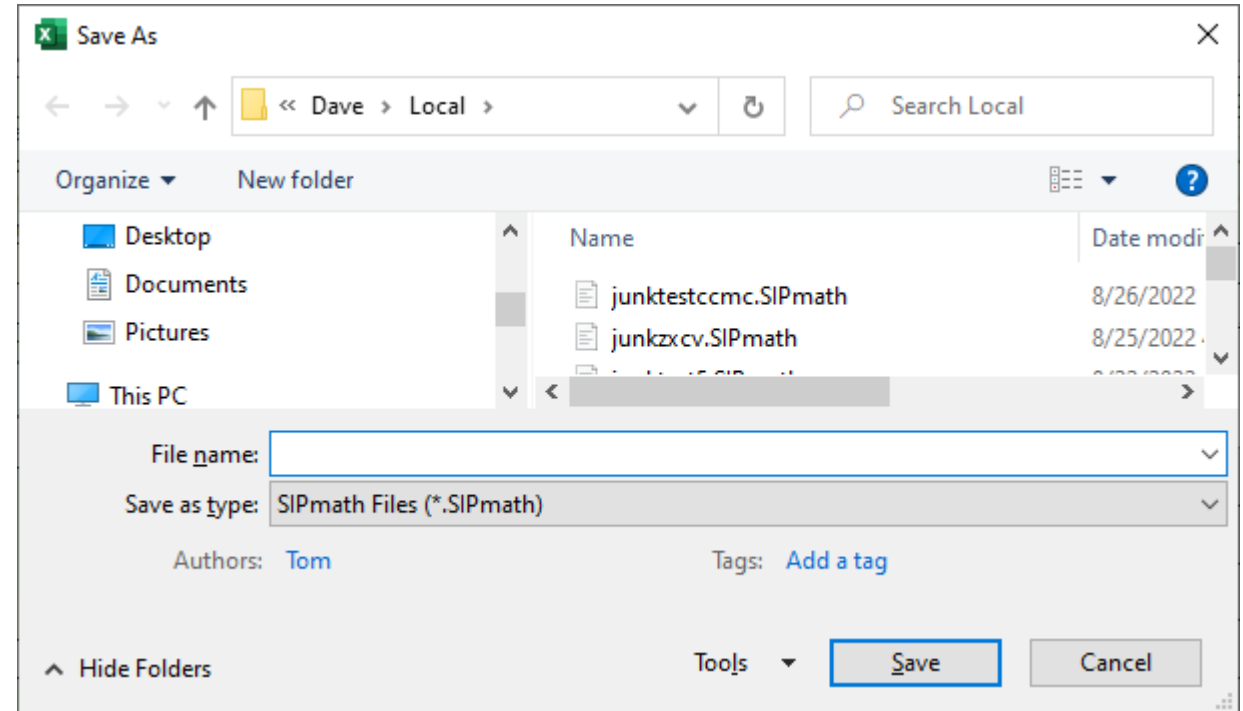


SIPmath mode, to File sub-mode

Button images and text will change depending on the last mode and sub-mode chosen

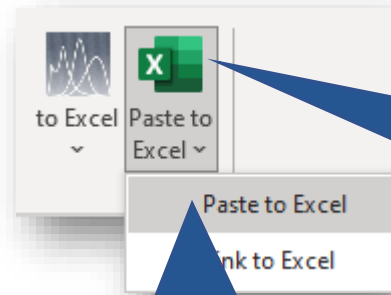
Create a single-SIP library and save it as a .SIPmath 3.0 JSON format file.

This is identical to the Clipboard sub-mode except the following dialog appears to save the file.





Excel mode, Paste sub-mode



Button images and text will change depending on the last mode and sub-mode chosen

Paste Metalog formulas into Excel without linking to Metalog template

Random cell to be used by Metalog distributions formula

Seeds for use by HDR random cell formula

Select random cell formula: HDR, Excel's RAND(), or a user formula (Last option leaves the formula in cell untouched)

Destination of Metalog distribution formula (you can select this before pressing Paste)

Check box and fill in destination for a PDF formula as well

Destination of sparkline for the distribution. The data for the sparkline will be copied to a tab named Sparkline Data.

Location of static CDF and PDF chart images

Paste Metalog Data

Distribution: A1

Enter PDF Dest: ☐

Sparkline Destination: A1

Charts Dest:

☒ External Random Cell ☒ Advanced HDR

Random Cell:

Start Variable ID (1-100 M): 1

Entity ID:

seed3 (1-100 M):

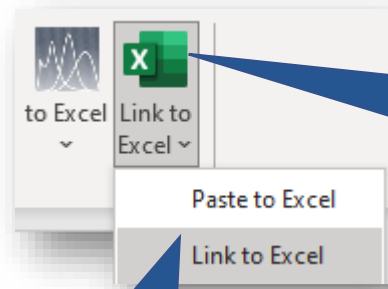
seed4 (1-10 M):

☒ HDR ☐ RAND ☐ User

OK Cancel



Excel mode, Link sub-mode



Button images and text will change depending on the last mode and sub-mode chosen

Links from the Metalog template sheet will be copied to this range, including links to sip name, bounds, and a-coefficients.

Add Metalog formulas into Excel linked to the Metalog template

NOTE: sparklines not directly available in this mode.

Select whether linked data will be in a row or a column.

This is otherwise identical to the Paste sub-mode except the metalog formulas and charts are all linked to the Metalog template sheet, and will reflect any changes to that template.

The 'Link Metalog Data' dialog box contains the following fields and options:

- Distribution: C1
- Enter PDF Dest: ☐
- Template Links:
- ☒ Data In Column ☐ Data in Row
- Charts Dest:
- ☒ External Random Cell ☒ Advanced HDR
- Random Cell:
- Start Variable ID (1-100 M): 1
- Entity ID:
- seed3 (1-100 M):
- seed4 (1-10 M):
- ☒ HDR ☐ RAND ☐ User
- OK Cancel

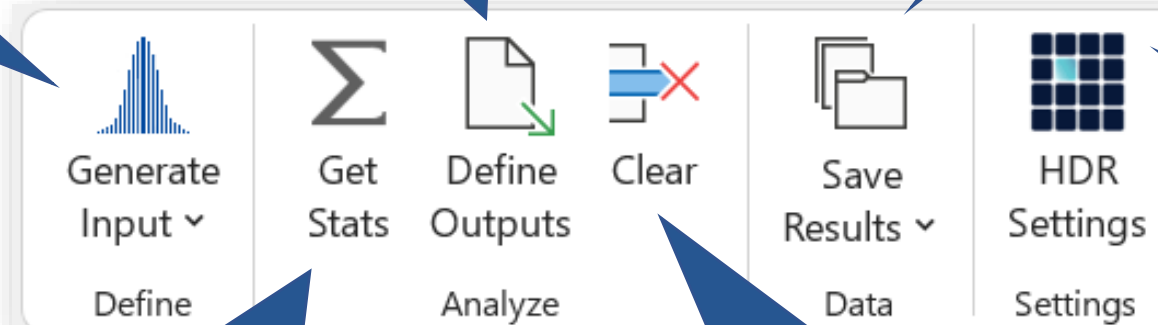


4. Monte Carlo Ribbon

a. Select from a list of formulas for any of several standard or custom distributions. (NOTE: this will also initialize the model if necessary.)

c. Defines selected cell as an output. (NOTE: this is usually done automatically when using the ChanceCalc tab. If necessary the model will also be automatically initialized.)

e. Saves current model data in SIPmath 2.0 XLS format, CSV format, or repeatedly saves data while changing a simulation parameter value.



b. Converts statistics formulas referring to output cells to refer to the corresponding output SIP range. (NOTE: this DOES NOT automatically initialize outputs.)

d. Un-defines the selected output and removes associated SIP data on PMTable sheet.

f. HDR Seed Attribute Names and Starting Var ID

Appendix: Enabling ActiveX Controls



ActiveX controls may need to be enabled for ChanceCalc to work properly. To enable ActiveX controls, go to the File tab and press the Options button at the bottom of the screen.

In the Options dialog, Select the Trust Center option.

In the Trust Center dialog, select the ActiveX Settings option.

Select either the third (shown here) or fourth ActiveX Settings option.

